

Bayesian Nonparametric Latent Class Models for Causal Inference in Cognitive Trajectories

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Motivating Study - Healthy Cognitive Aging

Why cognitive aging and dementia matter (Canada)

- Almost three-quarters of a million Canadians currently have dementia.
- Dementia is not normal aging, but changes in cognitive and daily function that diverge from norms **over time**.
- Prevention and risk-reduction efforts aim to **delay onset** and **slow progression** by targeting **modifiable risk factors**.

Clinical motivation

1. **Identify** actionable risk factors
2. **Quantify** how changing them may shift cognitive trajectories.

Modifiable Risk Factors for Dementia

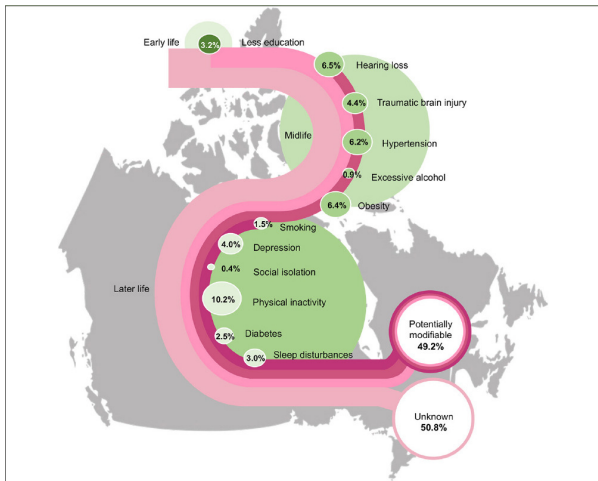


Figure 1: From Son et al. (2024)

- Son et al. estimated that 12 **potentially modifiable risk factors** may account for about **49.2%** of dementia cases in Canada.
- Key contributors include:
 - midlife excessive alcohol use and smoking
 - later-life physical inactivity
- Our CIHR project focuses on estimating causal effects of modifiable health behaviour risk factors.

Causal Question and Estimand

Causal question

If we intervene on a **modifiable risk factor**, how does the distribution of **latent cognitive-trajectory patterns** change?

Causal estimand. Let ζ_i^a be the trajectory class subject i would follow under risk-factor level a :

$$\psi_k(a) = P(\zeta_i^a = k), \quad \text{ATE}_k = \psi_k(1) - \psi_k(0).$$

Causal question

If we intervene on a **modifiable risk factor**, how does the distribution of **latent cognitive-trajectory patterns** change?

Two features make this hard:

- **Cognitive status is latent and multivariate-longitudinal.** It is **unobserved**; we measure it indirectly through repeated multiple cognitive tests over time. These outcomes **jointly** define heterogeneous latent patterns.
- **The number of trajectory patterns is unknown.** Conventional latent-class methods select the number of classes in a separate step, then condition on that single choice.

Existing Approach and Its Limitations

Existing method: Lanza et al. (2013)

- Latent class analysis groups subjects into unobserved classes from their responses. Lanza et al. (2013) use this to estimate the causal effect of an exposure on latent class membership with **cross-sectional** outcomes.
- **Key idea:** combine propensity score analysis with latent class analysis (a two-step approach)
 - Step 1. Estimate conventional propensity scores.
 - Step 2. Fit LCA on the outcome with treatment as an LCA covariate, adjusting for propensity scores via i) matching or ii) inverse probability weighting.
- Causal estimand: **Average Causal Effect**
 - The contrast in the distribution of outcome latent class membership if everyone in the population were exposed versus if no one were exposed.

Longitudinal extension as our baseline

- **Key idea for longitudinal extension:** combine propensity score analysis with latent class trajectory modeling (a two-step approach)
 - Step 1. Estimate conventional propensity scores.
 - Step 2. Fit **weighted latent class trajectory models** across candidate values of K , select the best model using BIC with a minimum class size $> 5\%$, and estimate the ATE from treatment-specific class membership probabilities.

Limitations we aim to address

This method selects and fixes K in a separate step, then conditions on that single choice. It is also estimated in two stages, so **uncertainty in K and class membership is not propagated to the causal estimate.**

⇒ This motivates a **full Bayesian framework.**

Building Blocks: Latent Class and BNP Models

Traditional Latent Class Model

Suppose the population is divided into K latent classes. For subject i ,

$$f(Y_i | X_i, Z_i, h_i) = \sum_{k=1}^K \pi_k(h_i) \phi(Y_i; \mu_{ik}, \Sigma_{ik}),$$

where $\pi_k(h_i)$ is the probability of allocation to class k .

Within each latent class, the longitudinal response is modeled by a class-specific LMM:

$$Y_i | \zeta_i = k = X_i \beta_k + Z_i b_i + \epsilon_i, \quad b_i | \zeta_i = k \sim \text{MVN}(0, \Lambda_k), \quad \epsilon_i \sim \text{MVN}(0, \sigma^2 I_{n_i}).$$

The class allocation probabilities are commonly modeled by multinomial logistic regression:

$$\pi_k(h_i) = \frac{\exp(\gamma_{k0} + h_i^\top \gamma_k)}{\sum_{\ell=1}^K \exp(\gamma_{\ell 0} + h_i^\top \gamma_\ell)}.$$

In standard LCMs, K is fixed in advance or selected. The final inference is then carried out **conditional on a single chosen K** , ignoring uncertainty in the number of classes.

Koo and Kim's BNP Extension

Koo and Kim (2020) proposed a Bayesian nonparametric LCM for longitudinal data. The finite mixture is replaced by an infinite mixture:

$$f(Y_i | X_i, Z_i, h_i) = \sum_{k=1}^{\infty} \pi_k(h_i) \phi(Y_i; \mu_{ik}, \Sigma_{ik}).$$

In practice, they use a large truncation level K and define predictor-dependent allocation probabilities:

$$\pi_k(h_i) = \frac{w_k \exp(h_i^\top \gamma_k)}{\sum_{\ell=1}^K w_\ell \exp(h_i^\top \gamma_\ell)}, \quad w_k \sim \text{Gamma}(\alpha/K, 1).$$

- w_k is the baseline weight for class k .
- γ_k controls how predictors affect allocation to class k .
- α controls the tendency to create more occupied classes.

This is the predictor-dependent normalized gamma process prior.

What the BNP Prior Adds

When predictor effects are removed, $\gamma_k = 0$,

$$\pi_k = \frac{w_k}{\sum_{\ell=1}^K w_{\ell}},$$

which gives a finite approximation to a Dirichlet-process mixture.

For posterior inference, Koo and Kim use a manually derived MCMC algorithm:

- Metropolis–Hastings updates for the predictor-dependent allocation parameters.
- Gibbs sampling updates for latent class memberships and class-specific LMM parameters.

Our starting point

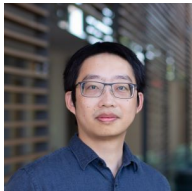
We build on this BNP latent class structure, then target causal effects on latent cognitive trajectory membership.

Proposed Method: BNP Latent Class Causal Model

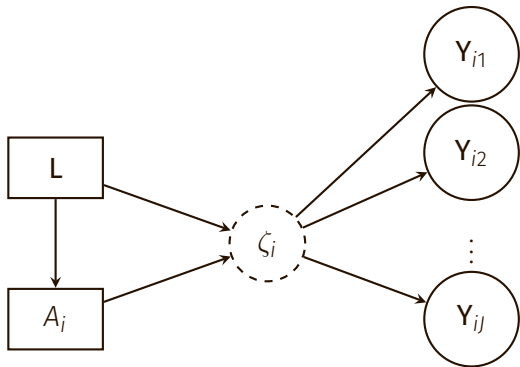
The proposed Bayesian approach

Project objectives

- We propose a **Bayesian nonparametric latent class causal framework** to estimate the effects of *modifiable risk factors* on *heterogeneous cognitive trajectories*, while accounting for *uncertainty in latent class membership* and *multiple cognitive domains*.
- Joint work by Emmett Peng (lead), Dr. Zihang Lu and Dr. Kuan Liu.



Bayesian nonparametric latent class model



Goal

Estimate how a **modifiable risk factor** shifts the **probability** of belonging to each latent cognitive trajectory class.

Setup

- L_i : baseline confounders.
- $A_i \in \{0, 1\}$: modifiable risk factor.
- ζ_i : latent cognitive trajectory class.
- Y_{ij} : multivariate cognitive outcomes at visit j ; manifest indicators of ζ_i .

Potential latent class

ζ_i^a : trajectory class under $A_i = a$.

Causal estimand

$$\psi_k(a) = P(\zeta_i^a = k), \quad \text{ATE}_k = \psi_k(1) - \psi_k(0).$$

Causal Assumptions and Joint Likelihood

Causal Assumptions:

- SUTVA and Positivity.
- Latent exchangeability: $A_i \perp\!\!\!\perp \zeta_i^a \mid L_i$.
- Conditional independence: $Y_i \perp\!\!\!\perp (A_i, L_i) \mid \zeta_i$.

The joint likelihood of the observed data is given as

$$\begin{aligned} L(\Lambda) &= \prod_{i=1}^n p(L_i \mid \Theta_L) p(A_i \mid L_i, \Theta_A) p(Y_i^{\text{obs}}, M_i \mid A_i, L_i, \mathcal{D}_i, \Lambda) \\ &\propto \prod_{i=1}^n p(L_i \mid \Theta_L) p(A_i \mid L_i, \Theta_A) \sum_{k=1}^{\infty} p(Y_i^{\text{obs}} \mid \zeta_i = k, \mathcal{D}_i, \Theta_Y) p(\zeta_i = k \mid A_i, L_i, \Theta_{\zeta}), \end{aligned}$$

where $\ell_{ik}(\Theta_Y) = \phi_{p_{ji}} \{ \text{vec}(Y_i^{\text{obs}}); \text{vec}(V_i B_k), R_{ik} \}$ and $p_k(G_i; \Theta_{\zeta}) = \frac{w_k \exp(G_i^{\top} c_k)}{\sum_{\ell=1}^K w_{\ell} \exp(G_i^{\top} c_{\ell})}$.

- Causal estimand (Average Potential Outcome):

$$E[\psi_k(a) \mid O_{1:n}] = \int \int p(\zeta_i = k \mid A_i = a, L^*, \Theta_\zeta) p(L^* \mid \Theta_L) p(\Lambda \mid O_{1:n}) dL^* d\Lambda.$$

- Causal estimation via parametric Bayesian g-computation (Oganisian, 2025), extending the g-formula (Robins, 1986)
 - Estimation algorithm in a nutshell (performed in Stan via *cmdstanr*)
 1. Specify and fit the joint likelihood with non-informative priors.
 2. Draw posterior samples $\Lambda^{(s)} = \{\Theta_Y^{(s)}, \Theta_L^{(s)}, \Theta_A^{(s)}, \Theta_\zeta^{(s)}\}$.
 3. G-computation: Generate synthetic draws of covariates $L^{(s,r)} \sim p(\cdot \mid \Theta_L^{(s)})$.
 4. Compute posterior predictive ATE:

$$\widehat{\text{ATE}}_k = \frac{1}{S \cdot M} \sum_{s=1}^S \sum_{r=1}^M \left[P(\zeta_i = k \mid A_i = 1, L^{(s,r)}, \Theta_\zeta^{(s)}) - P(\zeta_i = k \mid A_i = 0, L^{(s,r)}, \Theta_\zeta^{(s)}) \right]$$

Simulation Study and Results

Data-generating process

- $n = 300$ subjects, $J = 6$ visits at scheduled times $t \in \{0, 3, 6, 9, 12, 15\}$ months (jittered ± 0.5).
- $p = 3$ continuous cognitive outcomes; $K_{\text{true}} = 4$ latent trajectory classes.
- Class-specific cubic polynomial random-intercept model:

$$Y_{ij} = \mathbf{v}_{ij}^{\top} B_k + b_i + \varepsilon_{ij}, \quad b_i \sim \mathcal{N}(0, 0.10 I_p), \quad \varepsilon_{ij} \sim \mathcal{N}(0, 0.25 I_p).$$

- Binary treatment $A_i \sim \text{Bernoulli}(p_i)$ (logistic, 30% prevalence); confounders L_i : sex, education, age.

Class separation parameter δ

$B_k^{(\delta)} = \bar{B} + \delta(B_k - \bar{B})$; $\delta = 0$ collapses all classes to a common mean, $\delta = 1$ recovers the original class-specific curves. Baseline uses $\delta = 0.5$.

Simulation Scenarios

Four one-factor-at-a-time scenarios (each varied from the baseline):

Scenario	What changes	n	δ	Class props.	Null ATE
Baseline	—	300	0.50	25/25/25/25%	No
Null ATE	No treatment effect	300	0.50	25/25/25/25%	Yes
Low sep.	Weaker class separation	300	0.25	25/25/25/25%	No
Large n	Doubled sample size	600	0.50	25/25/25/25%	No

- 100 independent replications per scenario (SLURM array jobs on HPC).
- **Proposed:** BNP-LCM in Stan ($K_{\max} = 6$; 3,000 warmup + 1,000 sampling iterations, 3 chains).
- **Baseline:** IPW-weighted FlexMix, BIC-selected $K \in \{2, \dots, 6\}$, bootstrap CIs ($B = 50$).

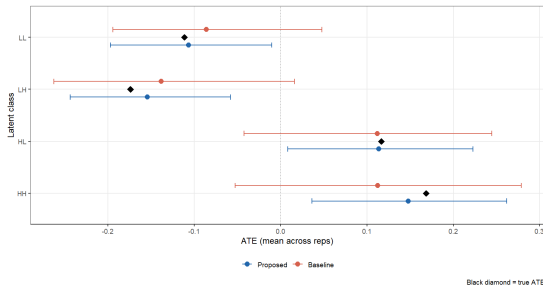
Results: Class Structure Recovery and ATE Estimation

Across 100 replications; RMSE and Coverage conditioned on K -correct replications.

Scenario	Proposed (BNP-LCM)			Baseline (IPW-FlexMix)		
	K -correct (%)	RMSE	Coverage (%)	K -correct (%)	RMSE	Coverage (%)
Baseline	89	0.052	95.8	62	0.083	95.6
Null ATE	85	0.053	95.0	52	0.059	97.6
Imbalance	79	0.050	94.9	56	0.081	96.9
Large n	90	0.035	95.6	60	0.068	95.8

- The proposed BNP model achieves **higher K -correct rates** and **lower RMSE** across all scenarios; both methods maintain **near-nominal coverage** when K is correctly identified.

Results: ATE Estimation (Baseline Scenario)



- **Proposed (blue)** estimates are closer to the true ATE (◆) across classes, indicating lower bias.
- **Baseline (red)** CIs are wider, suggesting less stable estimates from the concomitant model.
- Both methods show **near-nominal coverage**, but the proposed method gives **tighter intervals** through Bayesian *g*-computation.

Baseline scenario, *K*-correct replications only.
Proposed: 95% Bayesian credible intervals;
Baseline: 95% bootstrap CIs. ◆ = true ATE.

Real-Data Application: CLSA

Application: Canadian Longitudinal Study on Aging

Data source. Population-based cohort of middle-aged and older Canadian adults, with repeated cognitive assessments and rich lifestyle / clinical measures (Raina et al., 2019).

Analytic cohort ($n = 2,694$)

- Age ≥ 65 at baseline; complete cognitive data across **3 waves** (baseline, FU1, FU2).
- **Exposure** A_i : baseline depression (CES-D-10 ≥ 10).
- **Outcomes** Y_{ij} : 6 standardized tests — memory (REY_I, REY_II) and executive function (MAT, AF1, FAS, STP).

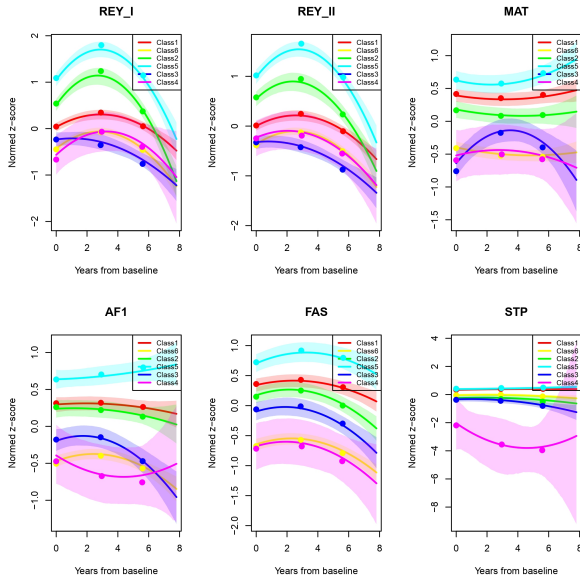
Baseline confounders L_i (Son et al., 2024)

- Sex, age, education, race/ethnicity, income.
- Modifiable risk factors: hearing, smoking, sleep, TBI, hypertension, alcohol, obesity, social isolation, physical activity (PASE), diabetes.

Causal estimand

Effect of baseline depression on the standardized probability of membership in each latent cognitive-trajectory class, $\text{ATE}_k = \psi_k(1) - \psi_k(0)$. Model fit once with truncation $K_{\max} = 10$.

Estimated latent cognitive-trajectory classes



- The BNP-LCM identified **6 active trajectory classes**
- Classes differ by **level** and **domain-specific change** over time, not just overall score.
- **Class 5**: high, stable function, low depression prevalence.
- **Class 3**: below-average, memory decline, highest depression prevalence.
- **Class 4**: rare, unstable severe-impairment pattern ($n = 30$).

Causal effect of baseline depression on trajectory membership

Class	Descriptive interpretation	Post. mean	2.5%	97.5%
1	Favourable executive fn., average memory	-0.129	-0.191	-0.064
2	Memory-preserved, weaker Stroop	0.021	-0.038	0.085
3	Lower trajectory, memory decline, high depression	0.093	0.031	0.160
4	Very small severe-impairment trajectory	0.004	-0.007	0.024
5	High cognitive function trajectory	-0.024	-0.073	0.029
6	Lower memory & executive-function trajectory	0.035	-0.038	0.113

- Baseline depression **lowers** the probability of the favourable **Class 1** ($ATE_1 = -0.129$) and **raises** that of the lower-performing **Class 3** ($ATE_3 = 0.093$); both credible intervals exclude 0.

⇒ Depression **shifts the population** away from favourable and toward lower-performing cognitive-aging trajectories.

Discussion and Future Directions

- Our proposed BNP-LCM approach demonstrates better performance across all scenario settings.
- **Future Research Directions:**
 - **Informative dropout and irregular visits:** The current model assumes covariate-dependent missingness (MAR). A joint trajectory-survival model could better handle informative dropout, such as death or dementia diagnosis, in cognitive aging cohorts.
 - **Autoregressive outcome model:** The current random-intercept model treats visits as conditionally independent given latent class and random effects. A Markov structure, $Y_{ij} \mid Y_{i,j-1}$, could capture short-term serial dependence beyond subject-level random effects.

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Structural recovery

- **K-correct rate:** % of replications recovering $K_{\text{true}} = 4$.
- **ARI:** Adjusted Rand Index (MAP-assigned vs. true class).
- **Macro-AUC:** mean one-vs-rest AUC across four classes (Mann-Whitney U).

Causal estimation

- **Bias:** $\widehat{\text{ATE}}_k - \text{ATE}_k^*$.
- **RMSE:** $\sqrt{\text{MSE}}$, class-averaged.
- **Coverage:** % of 95% CIs containing ATE_k^* .
- **Width:** mean 95% CI width.

True ATE oracle ATE_k^*

$\text{ATE}_k^* = E_L [P(\zeta_i^1 = k \mid L)] - E_L [P(\zeta_i^0 = k \mid L)]$, evaluated via g -formula on a synthetic population of $N = 10,000$ draws from the DGP.

MCMC Diagnostics: Active Classes and Label Switching

Determining K_{active}

- Stan returns soft class probabilities $P(\zeta_i = k \mid \cdot)^{(s)}$ per draw.
- Expected class size at draw s :
$$\hat{n}_k^{(s)} = \sum_{i=1}^n P(\zeta_i = k)^{(s)}.$$
- Class k declared **active** if
$$\hat{n}_k^{(s)} > n_{\min} = 20.$$
- Posterior distribution of
$$K_{\text{active}} = \sum_k \mathbf{1}[\hat{n}_k^{(s)} > n_{\min}]$$
 across draws summarises uncertainty in the number of components.

Resolving Label Switching

- Mixture posteriors are symmetric in class labels — MCMC chains can freely permute class indices across draws.
- **Manual PRA via Hungarian matching:**
for each draw s , find the column permutation $\sigma^{(s)}$ that maximises agreement with a fixed pivot draw:
$$\text{score}[j, k] = \sum_{i: \hat{\zeta}_i^{\text{pivot}} = j} P(\zeta_i = k)^{(s)},$$
solved exactly via the Hungarian algorithm.
- The same permutation is applied to B_k , $\log w_k$, and c_k before pooling draws for g -computation.

MCMC Diagnostics: Active Classes and Label Switching

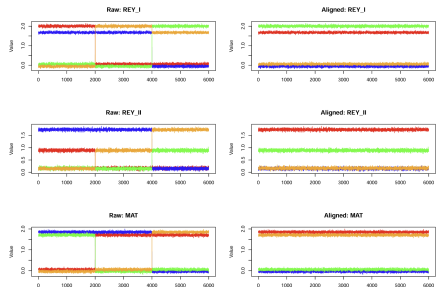


Figure 2: Trace plots for outcome means (θ) before and after alignment.

- **Raw Chains:** Exhibit label switching across iterations, where class indices permute between chains.
- **Aligned Chains:** Consistent class identification achieved by matching draws to a representation-based pivotal reference, enabling valid posterior inference and ATE estimation.
- **Label switching** corrected by solving the Linear Sum Assignment Problem to find optimal permutations (R package `label.switching`).